

OA Questions Format

➤ Recommended structure

- 1. Problem description** - Write the exact problem statement in simple and clear language.
- 2. Input format** - Mention how the input is given, line by line if needed.
- 3. Output format** - Mention the expected output clearly.
- 4. Constraints** - List constraints exactly as given or neatly rewritten.
- 5. Sample input** - Provide the sample input exactly as in the OA.
- 6. Sample output** - Provide the sample output exactly as in the OA.
- 7. Explanation** - Briefly explain the sample case and the logic behind the answer.
- 8. Tags** - Add company, topic, difficulty, and date/year **if available**.

➤ Template

Company: [Company Name]

Round: [OA / HackerRank / CodeSignal / Test]

Year: [YYYY]

Difficulty: [Easy / Medium / Hard]

Topic: [Arrays / Strings / DP / Graphs / etc.]

Problem Description:

[Write the question here.]

Input Format:

[Explain the input format.]

Output Format:

[Explain the output format.]

Constraints:

[Write all constraints here.]

Sample Input:

[Paste sample input here.]

Sample Output:

[Paste sample output here.]

Explanation:

[Explain the sample in 3-5 lines.]

➤ Example OA-style question

Problem Description:

You have an array **A** of **N** elements.

You are currently at **index 1** with a balance of **0**.

You can perform the following jump operation any number of times:

- Choose an index **j** such that **i < j**, where **i** is your current index.
- Jump from index **i** to index **j**.
- Your balance increases by:

$$A_j - j + i$$

Your goal is to find the **maximum possible balance** that can be obtained using any sequence of valid jumps.

Input Format:

The first line contains a single integer **T**, denoting the number of test cases.

For each test case:

- The first line contains a single integer **N**.
- The second line contains **N** integers:

$$A_1 \ A_2 \ \dots \ A_N$$

Output Format:

For each test case, print the maximum possible balance on a new line.

Constraints:

$$1 \leq T \leq 100$$

$$2 \leq N \leq 100$$

$$-100 \leq A_i \leq 100$$

Sample Input:

```
1
5
3 1 5 2 4
```

Sample Output:

```
6
```

Explanation:

We start at index 1 with balance 0.

One possible sequence of jumps is:

$1 \rightarrow 3 \rightarrow 5$

- For $1 \rightarrow 3$:
 $A_3 - 3 + 1$
 $= 5 - 3 + 1$
 $= 3$

Balance = 3

- For $3 \rightarrow 5$:
 $A_5 - 5 + 3$
 $= 4 - 5 + 3$

Balance = 2

Final balance: $3 + 2 = 5$